Question 1 :- Normalize the Following Table into 1NF .

MEMBER_ID	FIRST_NAME	LAST_NAME	HOBBIES
101	Jayson	Mark	Cricket, Swimming, Football
102	Ram	Ganesh	Swimming, Running, Music
103	Raj	Kishore	Dancing, Singing, Running

Solution:-

Normalize into 1NF

Create New Table

```
CREATE TABLE MEMBER_1NF (
   MEMBER_ID NUMBER,
   FIRST_NAME VARCHAR2(30),
   LAST_NAME VARCHAR2(30),
   HOBBY VARCHAR2(30)
);
```

Insert Values (Manual Method)

```
INSERT INTO MEMBER_1NF VALUES (101,'Jayson','Mark','Cricket');
INSERT INTO MEMBER_1NF VALUES (101,'Jayson','Mark','Swimming');
INSERT INTO MEMBER_1NF VALUES (101,'Jayson','Mark','Football');

INSERT INTO MEMBER_1NF VALUES (102,'Ram','Ganesh','Swimming');
INSERT INTO MEMBER_1NF VALUES (102,'Ram','Ganesh','Running');
INSERT INTO MEMBER_1NF VALUES (102,'Ram','Ganesh','Music');

INSERT INTO MEMBER_1NF VALUES (103,'Raj','Kishore','Dancing'); INSERT
INTO MEMBER_1NF VALUES (103,'Raj','Kishore','Singing');
INSERT INTO MEMBER_1NF VALUES (103,'Raj','Kishore','Running');

COMMIT;
```

Display 1NF Table

SELECT * FROM MEMBER_1NF;

Output

MEMBER_ID	FIRST_NAME	LAST_NAME	HOBBY
101	Jayson	Mark	Cricket
101	Jayson	Mark	Swimming
101	Jayson	Mark	Football
102	Ram	Ganesh	Swimming
102	Ram	Ganesh	Running
102	Ram	Ganesh	Music
103	Raj	Kishore	Dancing
103	Raj	Kishore	Singing
103	Raj	Kishore	Running

Question 2 :- Convert the following table into 2Nf.

EMPNO	TRAINING	TRAINING_DATE	DEPT
101	Oracle SQL	12-AUG-2015	TT
101	Java	21-AUG-2015	BU
102	Oracle SQL	18-SEP-2014	TT

Solution:-

Normalize into 2NF

Create Employee Table

```
CREATE TABLE EMPLOYEE (
   EMPNO NUMBER PRIMARY KEY,
   DEPT VARCHAR2(10)
);
```

Insert Employee Data

```
INSERT INTO EMPLOYEE VALUES (101,'TT');
INSERT INTO EMPLOYEE VALUES (102,'TT');

COMMIT;
```

Create Training Table

```
CREATE TABLE TRAINING (
   EMPNO NUMBER,
   TRAINING VARCHAR2(30),
   TRAINING_DATE DATE,
   PRIMARY KEY (EMPNO, TRAINING),
   FOREIGN KEY (EMPNO) REFERENCES EMPLOYEE(EMPNO)
);
```

Insert Training Data

```
INSERT INTO TRAINING
VALUES (101,'Oracle SQL',TO_DATE('12-AUG-2015','DD-MON-YYYY'));

INSERT INTO TRAINING
VALUES (101,'Java',TO_DATE('21-AUG-2015','DD-MON-YYYY'));

INSERT INTO TRAINING
VALUES (102,'Oracle SQL',TO_DATE('18-SEP-2014','DD-MON-YYYY'));
COMMIT;
```

Display Normalized Tables

1. Employee Table

```
SELECT * FROM EMPLOYEE;
```

Output

EMPNO	DEPT
101	TT
102	TT

2. Training Table

```
SELECT EMPNO,
    TRAINING,
    TO_CHAR(TRAINING_DATE,'DD-MON-YYYY') AS TRAINING_DATE
FROM TRAINING;
```

Output

EMPNO	TRAINING	TRAINING_DATE
101	Oracle SQL	12-AUG-2015
101	Java	21-AUG-2015
102	Oracle SQL	18-SEP-2014

Question 3: Normalize the following into 3NF .

MEMBER_ID	FIRST_NAME	LAST_NAME	SPORTS	FEES
101	Rajesh	Chand	Cricket	100
102	Jayesh	Raj	Hockey	80
103	Mark	Dorson	Football	90

Solution:-

Normalize into 3NF Create

Sports Table CREATE TABLE

```
SPORTS (
   SPORTS VARCHAR2(30) PRIMARY KEY,
   FEES NUMBER
);
```

Insert Sports Data

```
INSERT INTO SPORTS VALUES ('Cricket',100);
INSERT INTO SPORTS VALUES ('Hockey',80);
INSERT INTO SPORTS VALUES ('Football',90);

COMMIT;
```

Create New Member Table

```
CREATE TABLE MEMBER_NEW (
   MEMBER_ID NUMBER PRIMARY KEY,
   FIRST_NAME VARCHAR2(30),
   LAST_NAME VARCHAR2(30),
   SPORTS VARCHAR2(30),
   FOREIGN KEY (SPORTS) REFERENCES SPORTS(SPORTS)
);
```

Insert Member Data

```
INSERT INTO MEMBER_NEW VALUES (101,'Rajesh','Chand','Cricket');
INSERT INTO MEMBER_NEW VALUES (102,'Jayesh','Raj','Hockey');
INSERT INTO MEMBER_NEW VALUES (103,'Mark','Dorson','Football');

COMMIT;
```

Display Normalized Tables

Sports Table

```
SELECT * FROM SPORTS;
```

Output

SPORTS	FEES
Cricket	100
Hockey	80
Football	90

Member Table

SELECT * FROM MEMBER_NEW;

Output

MEMBER_ID	FIRST_NAME	LAST_NAME	SPORTS
101	Rajesh	Chand	Cricket
102	Jayesh	Raj	Hockey
103	Mark	Dorson	Football